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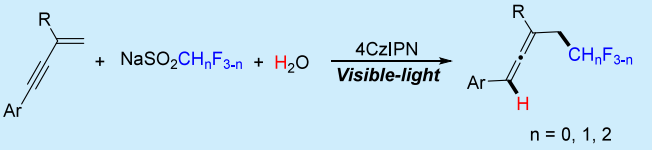
Visible-Light Photoredox-Catalyzed Radical-Polar Crossover 1,4-Hydrofluoromethylation of 1,3-Enynes

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ABSTRACT: We report herein a visible-light photoredox-catalyzed 1,4-hydrofluoromethylation of terminal-alkene-derived 1,3-enynes with sodium fluoromethylsulfinate, providing an effective protocol to access a diversity of di- and trisubstituted allenes under mild conditions. The synthetic utility of the present protocol was demonstrated by a large-scale reaction as well as the synthetic derivatization of the allene product.



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Fluoromethyls, such as CF_3, CHF_2, and CH_2F are important chemical groups in the field of medicinal chemistry[1,2] because of their high fat solubility, good metabolic stability, high electronegativity, and bioavailability. They have a wide range of applications in bioactive molecules.[3,4] Since the fluorinated compounds are relatively rare in nature, how to conveniently, gently, efficiently, and safely introduce fluorinated groups into various organic compounds to prepare special fluorinated compounds is an important research topic in synthetic chemistry.[5−7]

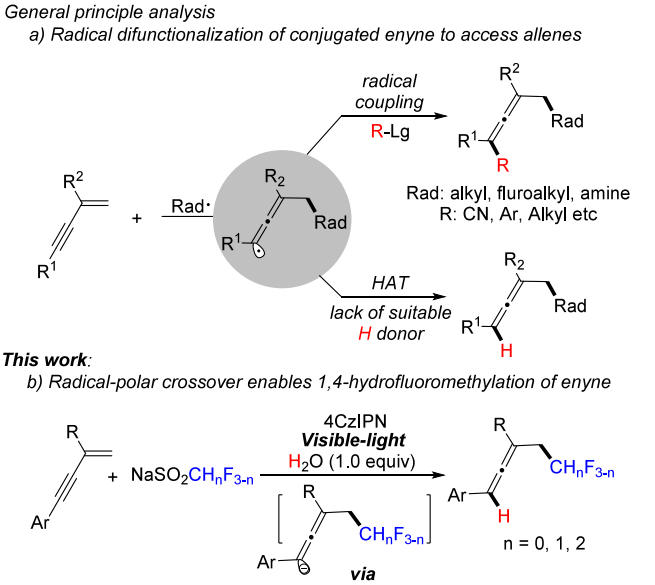
Allenes make up a class of organic molecules with unique physical and chemical properties. They are not only found as the key skeleton in numerous natural products and drug molecules but also serve as an important synthon in the organic synthesis of bioactive molecules.[8,9] So far, different strategies, including transition metal activation[10−12] and ionic 1,4-functionalization[13] of 1,3-enyne are well-developed to prepare allenes.

More recently, a variety of radical methods under thermal and photochemical reaction conditions have realized the installation of two functional groups on 1,3-enyne. The installation of a hydrogen atom onto the allenyl radical species would lead to hydrofunctionalization product formation, and it was a challenge due to the rare hydrogen atom donor for this hydrogen atom transfer (HAT) process (Scheme 1a).[14−18]

In the continuation of our photochemical functionalization project,[19,20] we herein developed a visible-light photoredox-catalyzed 1,4-hydrofluoromethylation of terminal-alkene-derived 1,3-enynes with sodium fluoromethylsulfinate, providing effective access to a diversity of di- and trisubstituted allene derivatives under transition-metal-free conditions (Scheme 1, b).

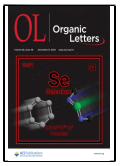
The present reaction development was started by testing the hydrotrifluoromethylation of 1,3-enynes (**1a**, 0.2 mmol) with $NaSO_2CF_3$ **2a** using 4CzIPN (2.0 mol %) as the photocatalyst and THF as the solvent, and Cs_2CO_3 (2.0 equiv) was used

Scheme 1. Radical Difunctionalization of 1,3-Enyne to Access Allenes in Principle and the Present Radical-Polar Crossover Enables 1,4-Hydrofluoromethylation of Enyne



under irradiation with a blue LED at room temperature for 12 h, giving the desired trisubstituted allenes **3a** in 60% isolated yield (entry 1, Table 1). Various solvents, including acetone,

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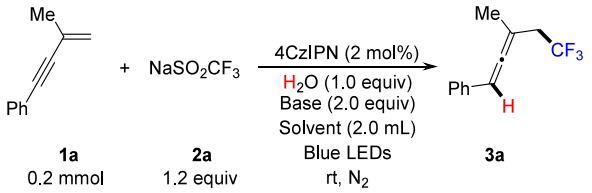


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DMF, DMSO, DCE, and MeCN were evaluated next, and MeCN gave **3a** in 88% isolated yield (entries 2−6, Table 1).

Table 1. Reaction Condition Optimization for the 1,4-Hydrotrifluoromethylation[a]



1a 0.2 mmol | **2a** 1.2 equiv | **3a**

entry	x	base	solvent	yield (%)[b]
1	1.5	Cs_2CO_3	THF	60
2	1.5	Cs_2CO_3	acetone	81
3	1.5	Cs_2CO_3	DMF	75
4	1.5	Cs_2CO_3	DMSO	46
5	1.5	Cs_2CO_3	DCE	9
6	1.5	Cs_2CO_3	MeCN	88
7	1.5	NaO^{t}Bu	MeCN	77
8	1.5	LiOH	MeCN	80
9	1.5	K_3PO_4	MeCN	65
10	1.5	DBU	MeCN	23
11[c]	1.5	Cs_2CO_3	MeCN	70
12	1.2	Cs_2CO_3	MeCN	88
13	1.2	Cs_2CO_3	MeCN/H_2O (v:v = 20:1)	38
14[d]	1.2	Cs_2CO_3	MeCN	NR
15[e]	1.2	Cs_2CO_3	MeCN	NR
16	1.2	-	MeCN	<5
17[f]	1.2	Cs_2CO_3	MeCN	NR

[a]Reaction conditions: **1a** (0.2 mmol), **2** (x equiv), 4CzIPN (2.0 mol %), base (2.0 equiv), solvent (2.0 mL), H_2O (3.6 μL) under nitrogen atmosphere, 12−24 h. [b]Isolated yield of **3a**. [c]1.0 mol % 4CzIPN was used. [d]The reaction was carried out under air atmosphere. [e]The reaction was carried out without the 4CzIPN. [f]The reaction was carried out under darkness.

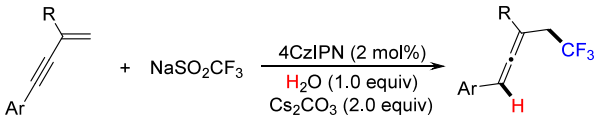
Different bases, such as NaO^{t}Bu, LiOH, K_3PO_4, and DBU were screened next, and no better result was obtained (entries 7−10, Table 1). The decrease in the amount of 4CzIPN led to a drop in the yield (entry 11, Table 1). It was surprising to observe that a decrease of the dosage of radical source **2a** did not affect the reaction, and 1.2 equiv of **2a** also gave **3a** in 88% yield (entry 12, Table 1). To our great surprise, when the mixture solvent of H_2O and MeCN was tested, the yield of **3a** dropped (entry 13, Table 1).

The visible light, photocatalyst, nitrogen atmosphere, and inorganic base were essential for this transformation, which was next confirmed (entries 14−17, Table 1).

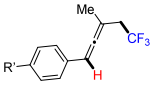
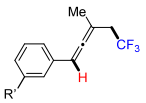
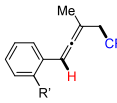
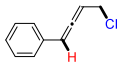
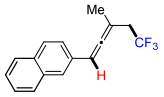
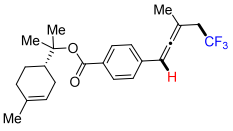
With the best reaction conditions established, different reactions of 1,3-enynes with sodium trifluoromethanesulfinate **2a** were carried out to explore the substrate scope, and the results are listed in Table 2. In general, 1,3-enynes bearing halogen atoms (Br, Cl, and F) at the *para*-position of the phenyl reacted smoothly under the standard conditions and provided the desired products **3b**−**3d** in 66−98% yields (entries 2−4, Table 2). Allene with *para*-trifluoromethyl substituent **3e** was obtained in 95% yield (entry 5, Table 2).

Interestingly, electron-donating groups, such as OMe and OEt, at the *para*-position of the phenyl reduced yields, and the corresponding products **3f** and **3g** were obtained in 40% and 43% yields, respectively (entries 6 and 7, Table 2). Meanwhile,

Table 2. Enyne Scope for the 1,4-Hydrotrifluoromethylation[a]



1 0.2 mmol | **2a** 1.2 equiv | **3**

entry	product	**3**	Yield (%)[b]
1		R′ = H, **3a**	88
2		R′ = Br, **3b**	98
3		R′ = Cl, **3c**	90
4		R′ = F, **3d**	66
5		R′ = CF_3, **3e**	95
6		R′ = OMe, **3f**	40
7		R′ = OEt, **3g**	43
8		R′ = Ph, **3h**	84
9		R′ = CN, **3i**	70
10		R′ = CO_2Me, **3j**	67
11		R′ = CHO, **3k**	51
12		R′ = NO_2, **3l**	46
13		R′ = OMe, **3m**	67
14		R′ = Me, **3n**	61
15		R′ = CF_3, **3o**	80
16		R′ = Br, **3p**	72
17		R′ = Cl, **3q**	97
18		R′ = F, **3r**	81
19		R′ = Br, **3s**	87
20		R′ = Cl, **3t**	91
21		R′ = F, **3u**	73
22		R′ = CF_3, **3v**	71
23		R′ = OMe, **3w**	72
24		**3x**	68
25		**3y**	87
26		**3z**	51

[a]Reaction conditions: **1** (0.2 mmol), **2a** (0.24 mmol), 4CzIPN (2.0 mol %), Cs_2CO_3 (2.0 equiv), MeCN (2.0 mL), H_2O (3.6 μL) under nitrogen atmosphere, 12−24 h. [b]Isolated yield of **3a** to **3z**.

the yields were increased with the switch of electron-donor groups from the *para*-position to *ortho*- and *meta*-positions (entries 13, 14, and 23, Table 2).

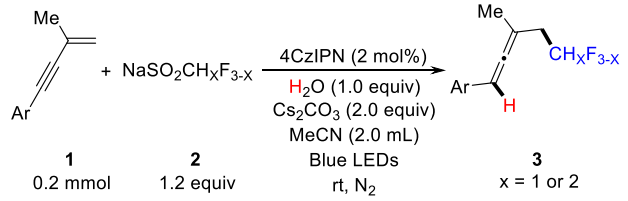
1,3-Enynes bearing different functional groups, such as phenyl, cyano, ester, and formyl, were all compatible, delivering the desired products **3h**, **3i**, **3j**, and **3k** in good yields. Halogen atoms at the *meta*- and *ortho*-positions of the benzene ring also delivered the desired allene products **3p**−**3r** and **3s**−**3u** in moderate to excellent yields. Disubstituted allene **3x** was also successfully obtained in 68% yield. The naphthalene ring containing allene product **3y** was obtained in 87% yield. 1,3-Enyne bearing a structurally complex ester group on the benzene ring was also well tolerated, giving desired product **3z** in 51% yield (entries 24−26, Table 2).

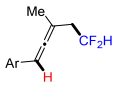
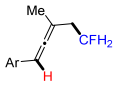
The scope of fluoromethyl radical sources for the present transformation was carried out further, and $NaSO_2CHF_2$ as a radical source was evaluated. A wide range of aryl-substituted

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1,3-enynes could be effectively applied to these reaction conditions. As shown in Table 3, electron-donating functional

Table 3. Reaction Scope for the 1,4-Hydrofluoromethylation[a]



entry	products	3	Yield (%)[b]
1		Ar = 2-$CF_3C_6H_4$, **3aa**	90
2		Ar = 2-BrC_6H_4, **3ab**	85
3		Ar = 2-FC_6H_4, **3ac**	83
4		Ar = 3-$OMeC_6H_4$, **3ad**	49
5		Ar = 3-MeC_6H_4, **3ae**	43
6		Ar = 4-ClC_6H_4, **3af**	65
7		Ar = 4-PhC_6H_4, **3ag**	76
8		Ar = 2-Naphthyl, **3ah**	70
9		Ar = 2-ClC_6H_4, **3ai**	68
10		Ar = 2-$CF_3C_6H_4$, **3aj**	82
11		Ar = 4-CNC_6H_4, **3ak**	51
12		Ar = 4-BrC_6H_4, **3al**	56
13		Ar = 2-Naphthyl, **3am**	61

[a]Reaction conditions: **1** (0.2 mmol), **2** (0.24 mmol), 4CzIPN (2.0 mol %), Cs_2CO_3 (2.0 equiv), MeCN (2.0 mL), H_2O (3.6 μL) under nitrogen atmosphere, 12−24 h, and isolated yield of **3aa**−**3am**.

groups, including methyl and methoxy, as well as electron-withdrawing functionalities such as halogen (Br, Cl, and F), trifluoromethyl, and phenyl, were all compatible with the standard reaction conditions, giving **3aa**−**3ag** in good yields (entries 1−7, Table 3).

Significantly, this protocol was also compatible with $NaSO_2CH_2F$. For example, 1,3-enynes bearing trifluoromethyl, cyano, and halogen substituted on the phenyl ring all gave the desired products **3ai**−**3al** in good yields (entries 9−12, Table 3). Furthermore, naphthalene-containing 1,3-enynes were suitable for this 1,4-hydrofluoromethylation protocol, and the desired products **3ah** and **3am** were obtained in 70% and 61% isolated yields, respectively (entries 8 and 13, Table 3).

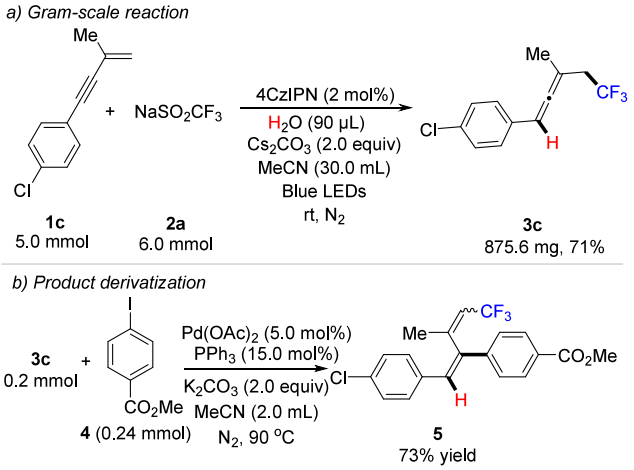
It should be mentioned that, although some products were obtained in moderate isolated yields, possible dimerization of 1,3-enynes was completely suppressed, and low reaction conversion was observed in those cases.

A gram-scale reaction was carried out next, and 5.0 mmol **1c** was successfully subjected to this reaction, giving **3c** in 71% yield (Scheme 2a). The synthetic applicability of this protocol was further demonstrated by the product derivatization, and the arylation led to the formation of 1,3-diene **5** in 73% yield (Scheme 2b).[21]

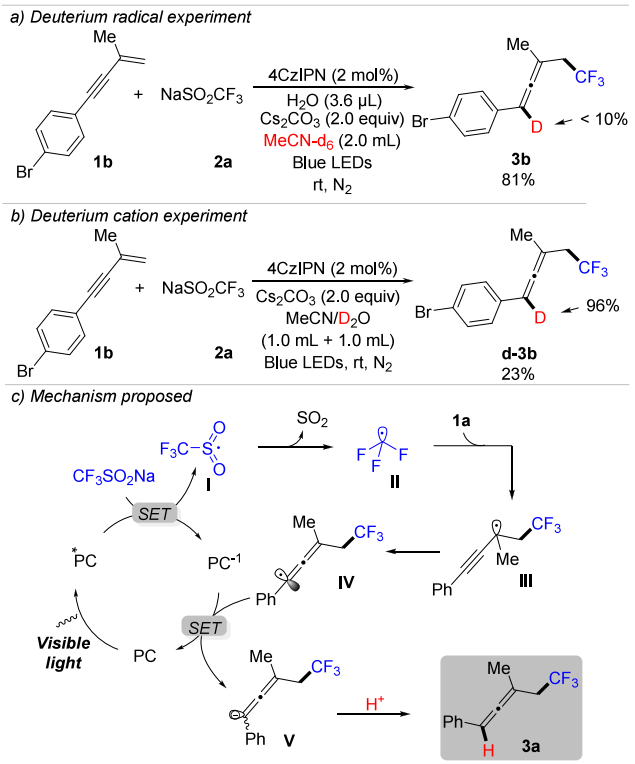
To understand more detailed information on the reaction mechanism for this 1,4-hydrofluoromethylation protocol, parallel deuterium radical and deuterium cation experiments were carried out, and the results suggested that the hydrogen atom of the sp^2 C−H bond in the allene came from the trace amount of water rather than the MeCN solvent (Scheme 3a,b).

Based on the above reaction development and previous reports, a possible reaction mechanism for the present 1,4-hydrofluoromethylation reaction was proposed in Scheme 3c.

Scheme 2. Gram-Scale Reaction and Product Derivatization



Scheme 3. Mechanistic Studies and Mechanism Proposed



The single electron oxidation of sodium trifluoromethanesulfonate by the excited species of the photocatalyst would generate trifluoromethylsulfonyl radical species **I** and rapidly remove SO_2 for formation of the trifluoromethyl radical **II**. The addition of trifluoromethyl radical with 1,3-enyne **1a** to generate propargyl radical **III**, which then rapidly forms allenyl radical **IV** via resonance, forms anion species **V** under the single electron reduction of PC^{-1}. And the protonation of species **V** by water would lead to the desired product **3a** formation.

The use of inorganic base, as well as a trace amount of water, were important for this transformation, probably benefiting from the suppression of generated sulfuric acid.

In conclusion, we describe a visible-light photoredox-catalyzed 1,4-hydrofluoromethylation of 1,3-enyne. The use

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of sodium fluoromethylsulfinate as the radical source and a trace amount of water as the hydrogen atom source allows this approach for a diversity of di- and trisubstituted allenes. The synthetic utility of this protocol is demonstrated by a large-scale reaction and derivatization reaction.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.4c04089.

Methods description, synthesis and characterization of products, and NMR spectra (PDF)

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Notes

The authors declare no competing financial interest.

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